

# Harsath Kaliappa Gounder Thangavel

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## Education

### York University

Bachelor of Science in Computer Science (Honours)

Toronto, Canada

January 2020 – December 2024

## Experiences

### Software Engineer

Privex Inc.

October 2023 – January 2024

London, United Kingdom

- Developed an AI-driven support assistant using Retrieval-Augmented Generation (RAG) on a self-hosted LLM, reducing staff workload by handling common inquiries related to technical issues, products, and services.
- Implemented REST APIs to interface with the LLM and integrated Discord APIs to enable customer queries and on-demand data updates by staff.
- Designed data ingestion and management workflows to process and structure context data from sources such as website documentation, bulk files, support tickets, and Discord announcements.
- Built and maintained infrastructure for efficient data storage and retrieval, using MySQL for chunked data storage and ChromaDB for optimized embedding retrieval.
- Created systemd service files and monitoring scripts to ensure system reliability, maximize uptime, and automate error recovery.
- Tech Stack:** Python, Flask, ChromaDB, Langchain, MySQL, Linux

### Software Engineer

CogniPet AG

April 2019 – December 2019

Zürich, Switzerland

- Developed Pet Breed Detection system using the Xception neural network and transfer learning in Keras, achieving 94% accuracy across 120 dog breeds.
- Implemented REST APIs for model inference and integrated them into CogniPet's Android and iOS apps using AWS EC2.
- Built secure data management workflows for user-uploaded pet images in AWS S3, ensuring GDPR compliance.
- Explored video-based techniques for pet facial recognition by collecting and analyzing multi-frame data from dogs to improve feature extraction methods.
- Tech Stack:** Python, Flask, Keras, AWS (EC2, S3)

### Founder and Developer

TaffyHub

August 2017 – July 2019

Coimbatore, India

- Developed and launched school-specific social networking platform for fellow students, reaching over 500+ registered users with 150+ daily active users.
- Implemented key features, including news feed, friend recommender system, content sharing system, engagement system (likes, comments, tags), Q&A system, and premium membership tier offering exclusive perks.
- Built the backend using PHP and MySQL, managing approximately 100 posts per day efficiently.
- Configured Nginx reverse proxies for load balancing and media file caching, and utilized Memcached to optimize query performance.
- Managed the full DevOps lifecycle, including server setup, automated backups, and system maintenance.
- Generated \$100/month in revenue through Google AdSense and premium memberships.
- Tech Stack:** PHP, JavaScript, MySQL, Memcached, Linux (rsync, cron)

## Projects

### [Jix Interpreter](#) | C Interpreter

- Developed an interpreter for a dynamically typed programming language.
- Implemented fault-tolerant parsing and error handling, arrays, function pointers, loops, nested blocks, recursion, built-in functions, conditional statements, and control flow statements.
- Implemented basic functional programming support, including first-class and higher-order functions.
- Tested using automated tests with CI/CD pipelines.
- Tech Stack:** C, CMake, GitHub Actions

### [Prism Coverage](#) | Java Code Coverage Engine

- Developed a code coverage engine using ANTLR4 for a programming language.
- Implemented control flow coverage (statement, condition, path) and data-flow coverage (All-Defs, All-DU-Paths, All-P-Uses).
- Implemented color-coded HTML report generation for the coverage engine.
- Tech Stack:** Java, Maven, JUnit, CI/CD, GitHub Actions

## Technical Skills

**Languages:** Python, JavaScript, TypeScript, Java, C++, C, SQL, PHP, Perl, Bash, HTML/CSS  
**Technologies:** Linux, PostgreSQL, Docker, Kubernetes, MongoDB, Memcached, Redis, AWS, Neo4j  
**Frameworks:** Node.js, React, Flask, Django, Langchain  
**Tools:** Git, CI/CD, CMake, Maven, Linux Containers (LXC/LXD)